

and setting the threshold value for determining when the signal is high and when it is low. While the concept seems simple enough, when setting up protocol decode for a serial bus with three, four or five signals, the task becomes more complex than originally anticipated. If decode is set up for multiple simultaneous serial buses, the task becomes even trickier.

One innovative feature that some vendors offer is 'auto setup' for decode. After the user assigns channels, the auto setup works a bit like autoscale. It determines the correct threshold level for each signal and scales the timebase appropriately. This feature is particularly effective for users who don't make decode measurements frequently.

3 How useful is protocol decode display?

Decode display can vary from scope to scope to a high degree. Display protocol decode for the bus you are interested in on the scope you are considering.

Annotating waveforms with decode is the most common method of displaying protocol packets. Packet content is aligned in time with parametric signal detail. Some vendors colourise packets to make it easier to determine packet sequence even with larger timebase settings when packet detail is too compressed to see detail. Some vendors decode the entire acquisition memory, while others decode only what is on the screen. Decoding only on-screen info can be problematic. If the entire packet is not displayed on the screen, the scope would not decode any of the packets, or worse, may decode a partially displayed packet incorrectly.

Most vendors also offer a lister that will display decode of sequential packets. Listings let users see the flow of packets in a more condensed format (see Fig. 1). Unlike decode in waveform areas, listings show packet detail independent of timebase settings. Listing detail can vary greatly from vendor to vendor and scope to scope.

When evaluating protocol decode

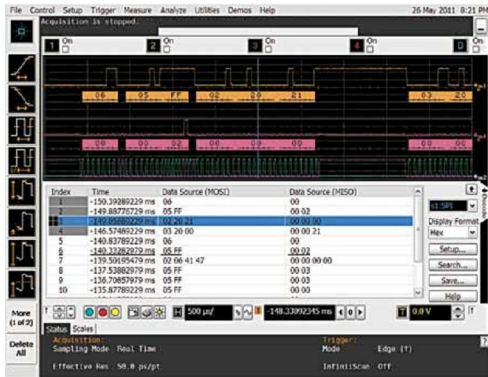


Fig. 2: Time-aligned markers track in the listing when moved in the waveform area, or conversely tracks in the waveform area when a new row of the lister is highlighted as shown in this SPI decode example

on scopes that incorporate a lister, ensure that your vendor provides time alignment between each row in the lister and signals in the waveform display. With time alignment, users can move between physical layer and packet layer quickly and confidently (see Fig. 2). Some vendors provide listings with minimal or no time alignment with signal detail, which makes debug more challenging.

4 What type of packet triggering is incorporated in protocol application?

Scope vendors typically bundle packet-based triggers with each decode application. These triggers can be implemented in software or hardware. Knowing this level of detail is important if the user plans triggering on infrequent events.

Hardware-based serial packet triggers are typically implemented in an FPGA, and run in real time. The vendor implements a real-time state machine that tracks incoming packet content and when the specified condition is met, this hardware interacts with trigger circuitry of the scope.

Hardware-based protocol triggers will absolutely trigger the scope on a specified packet no matter how infrequent the event occurs.

Software-based serial packet triggers can be developed more easily. They accomplish the task using post-processing software and hence are slower than hardware-based protocol triggers. After each acquisition, the software searches the acquisition record for the specified packet. If the specified condition is found, the scope displays the acquired memory buffer including both signal and packet detail. Alternatively, if the software searches the acquired memory and does not find the specified trigger condition, the scope discards the acquisition and acquires a new acquisition. Due to the high amount of processing after each acquisition, software-based triggering has significant dead time between acquisitions and is extremely likely to miss trigger conditions that occur infrequently. As memory depth increases, so does processing time, thus significantly increasing dead time between acquisitions.

So pull up a serial trigger on your vendor's scope for the protocol you